

Spin-Polarized Fusion × Adjoint Magnet Shielding

A working primer on the directional source, the importance method, and where the project goes

Internal note — QH/QA stellarator neutronics — compiled August 5, 2026

The one-paragraph version. We are building two coupled capabilities. (1) A *spin-polarized fusion* (SPF) neutron source for OpenMC whose *birth directions* are steered by the fuel polarization relative to the local magnetic field $\hat{\mathbf{B}}$ — physics from Schwartz (2025). (2) An *adjoint importance / contributon* method that identifies, for a given magnet, *which* neutrons and *which* shield locations set its radiation load — and then re-partitions blanket material (breeder ↔ shield, at fixed machine envelope) to protect it. The scientific payoff is a distinction nobody has published: *where the responsible neutrons are born* (attribution) is *not where blocking them helps* (worth). Everything below builds that claim from the physics up.

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1 The project in one page

Motivation. In a compact stellarator reactor the superconducting coils are the most neutron-sensitive, most expensive, and least replaceable components. Fusion neutrons born in the plasma stream outward; a fraction reach the coils and cause heating, insulation damage, and flux that quenches superconductivity. Two levers reduce that load: change *which* neutrons are born going *where* (the SPF source), and change *where* the shield sits (nonuniform blanket design). This project develops both and—crucially—the adjoint machinery that tells you *where* to spend either lever.

Two papers. The work splits cleanly:

- **Paper A — SPF directional source × adjoint magnet importance.** Couple the polarization-steered birth distribution to the coil importance map: does aligning the emission lobe away from a coil measurably lower its load? Strongest novelty (no one has crossed SPF directionality with magnet adjoints).
- **Paper B — adjoint-informed nonuniform magnet shielding.** At fixed outer envelope, trade tritium-breeder thickness for neutron-shield thickness, *locally*, guided by the adjoint. The headline is not the optimizer (others have nonuniform-blanket tools) but the *experiment*: attribution ≠ actionability, validated by finite difference.

Read this framing first. The tooling (compiled source, ParaStell geometry, random-ray adjoint) is largely prior art we build *on*. Our defensible contribution is a *scientific result*: the importance map everyone uses as a placement lever is the wrong quantity, and here is the right one, validated on hardware.

2 The SPF directional source: physics

2.1 Why polarization steers neutrons

In D–T fusion the dominant reaction proceeds through a $J^\pi = 3/2^+$ resonance. When the deuteron and triton spins are *polarized* relative to a quantization axis (the local field $\hat{\mathbf{B}}$), the angular distribution of the emitted neutron is no longer isotropic: it acquires a $\sin^2 \theta$ or $\cos^2 \theta$ character in θ , the angle between the neutron direction $\hat{\mathbf{u}}$ and $\hat{\mathbf{B}}$. Aligning the fuel thus both changes the *total* rate and *steers* the emission — the physics we exploit.

2.2 Mode parameterization (Schwartz 2025, Eq. 1)

Let deuteron spin-projection fractions be d_+, d_0, d_- and triton fractions t_+, t_- . Group the collision channels into three mode fractions:

$$a = d_+ t_+ + d_- t_-, \quad b = d_0, \quad c = d_+ t_- + d_- t_+, \quad a + b + c = 1. \quad (1)$$

Unpolarized fuel is $a = b = c = \frac{1}{3}$. Note b is simply the deuteron transverse ($m = 0$) fraction. This (a, b, c) grouping is *not* the naive “ $S = 3/2$ aligned / mixed / $S = 1/2$ ” split, and a common error (mode c “is isotropic”) is wrong—see below.

2.3 Differential cross section (Schwartz 2025, Eq. 2) — authoritative

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\sigma_0}{2\pi} \left[\frac{3}{4} a \sin^2 \theta + \left(\frac{2}{3} b + \frac{1}{3} c \right) \left(\frac{1}{4} + \frac{3}{4} \cos^2 \theta \right) \right]} \quad (2)$$

with $\theta = \angle(\hat{\mathbf{u}}, \hat{\mathbf{B}})$ and σ_0 the $S = 3/2$ combined-spin cross section. Integrating (2) over 4π (do it in sympy; uses $\int \sin^2 \theta \, d\Omega = 8\pi/3$ and $\int (\frac{1}{4} + \frac{3}{4} \cos^2 \theta) \, d\Omega = 2\pi$):

$$\sigma_{\text{tot}} = \sigma_0 \left[a + \frac{2}{3} b + \frac{1}{3} c \right]. \quad (3)$$

Mode	(a, b, c)	$\sigma_{\text{tot}}/\sigma_0$	Angular shape
Unpolarized	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	2/3	isotropic
A	$(1, 0, 0)$	1 (+50%)	$\sin^2 \theta$ — peaks $\perp \hat{\mathbf{B}}$, zero along $\hat{\mathbf{B}}$
B	$(0, 1, 0)$	2/3	$\frac{1}{4} + \frac{3}{4} \cos^2 \theta$ — peaks $\parallel \hat{\mathbf{B}}$
C	$(0, 0, 1)$	1/3 (half)	<i>same shape as B</i> , not isotropic

Table 1: Corner values — your oracle. Verify each from (3) before trusting any sampler. Non-polarized is $2\sigma_0/3$, *not* σ_0 .

The correction to remember. Setting $a = 0$ in (2), both the b and c terms multiply the *same* function $\frac{1}{4} + \frac{3}{4} \cos^2 \theta$. So **B and C share directionality** and differ *only* in total rate (2/3 vs. 1/3). Mode C is *not* isotropic. Pure A ($\propto \sin^2 \theta$) is the only qualitatively different lobe: it emits *perpendicular* to $\hat{\mathbf{B}}$ and vanishes along it.

2.4 Separate direction from rate

Because the b, c terms share an angular function, the birth-*direction* pdf depends on only two effective weights:

$$w_{\perp} = \frac{3}{4} a \quad (\text{the } \sin^2 \theta \text{ piece}), \quad w_{\parallel} = \frac{2}{3} b + \frac{1}{3} c \quad (\text{the } \frac{1}{4} + \frac{3}{4} \cos^2 \theta \text{ piece}). \quad (4)$$

The unnormalized intensity is $I(\theta) = w_{\perp} \sin^2 \theta + w_{\parallel} (\frac{1}{4} + \frac{3}{4} \cos^2 \theta)$, azimuthally symmetric about $\hat{\mathbf{B}}$; the pdf is $p(\theta, \phi) = I / \int I \, d\Omega$ with $\phi \sim U[0, 2\pi)$. The overall total-rate factor

$$\eta = a + \frac{2}{3} b + \frac{1}{3} c \quad (5)$$

is a *separate scalar* that multiplies the spatial source strength.

Design decision that mirrors Schwartz. *Direction* needs only $(w_{\perp}, w_{\parallel})$; *rate* is the scalar η . Keep them apart in code: A-mode and C-mode produce different *numbers* of neutrons but each produces correctly *shaped* directions. For pure-shape verification you normalize η out; for absolute wall loads it multiplies the strength.

2.5 The pdfs you actually sample

In $\mu = \cos \theta$ (absorbing the $\sin \theta$ Jacobian), the two normalized shapes on $[-1, 1]$ are

$$p_A(\mu) = \frac{3}{4}(1 - \mu^2) \quad (\text{vanishes at } \mu = \pm 1), \quad p_{BC}(\mu) = \frac{1}{4} + \frac{3}{4}\mu^2 \quad (\text{peaks at } \mu = \pm 1). \quad (6)$$

Energy is decoupled to leading order: emit monoenergetic 14.06 MeV for the analytic checks, or a Ballabio-broadened Gaussian (≈ 14.08 MeV, width $\approx 177\sqrt{T_i[\text{keV}]}$ keV) behind a flag.

3 The SPF directional source: computation

3.1 Stratified angular sampling (exact)

Sample the mixture by *stratifying* over the two shapes:

1. With probability $P_\perp = W_\perp / (W_\perp + W_\parallel)$ pick the $\sin^2$ shape, else the $\frac{1}{4} + \frac{3}{4}\cos^2$ shape. Here $W_\perp = w_\perp \cdot \frac{8\pi}{3}$ and $W_\parallel = w_\parallel \cdot 2\pi$ are the *solid-angle-integrated* weights (derive the constants; do not paste them).
2. Draw $\mu = \cos \theta$ from the chosen shape by **inverse-CDF**: each CDF is a cubic in μ ; solve the depressed cubic (trig method) and pick the root in $[-1, 1]$. Cross-check with a **rejection** sampler (envelope $\propto \sin \theta$) in a unit test — “inverse-CDF vs. rejection” catches a wrong root selection.
3. $\phi \sim U[0, 2\pi)$.

This yields exact samples for *any* (a, b, c) and makes per-mode χ^2 verification trivial.

3.2 Local $\rightarrow$ global frame rotation (no gimbal trap)

Sample in the $\hat{\mathbf{B}}$ -aligned local frame ($\hat{z}_{\text{loc}} \parallel \hat{\mathbf{B}}$), then rotate. Do *not* build the basis from $\hat{\mathbf{B}} \times \hat{z}_{\text{world}}$ (degenerate when $\hat{\mathbf{B}} \parallel \hat{z}$). Use least-aligned reference axis + Gram–Schmidt:

$$\text{ref} = \arg \min_{\hat{e} \in \{\hat{x}, \hat{y}, \hat{z}\}} |\hat{\mathbf{B}} \cdot \hat{e}|, \quad \hat{x}_{\text{loc}} = \widehat{\text{ref} - (\text{ref} \cdot \hat{\mathbf{B}})\hat{\mathbf{B}}}, \quad \hat{y}_{\text{loc}} = \hat{\mathbf{B}} \times \hat{x}_{\text{loc}}.$$

Assert $|\hat{\mathbf{u}}| - 1 < 10^{-12}$. Unit-test the degenerate $\hat{\mathbf{B}} = \pm \hat{z}$ branch explicitly.

3.3 OpenMC integration and hard rules

- Implemented as a `CompiledSource` (`PolarizedFusionSource` / `StellaratorSource`) whose `sample(seed)` returns a `SourceSite` $(\mathbf{r}, \hat{\mathbf{u}}, E, \text{wgt}, \dots)$.
- **RNG**: only `openmc::prn(seed)`. Never `rand()` or `<random>`.
- **Thread safety**: `sample()` runs concurrently per-thread; *any* mutable member is a data race. All state is set in the constructor and `const` thereafter.
- **Gate on invariants (assert), warn only on user input**. Renormalizing $a + b + c$ warns; “selection probabilities sum to 1” asserts.

Verification ladder. (T1) direction sampler alone: χ^2 of sampled $\cos \theta$ vs. $p(\mu)$ for each mode; rotation isotropy; inverse-CDF vs. rejection KS test. (T2) full-stack neutron-wall-load vs. the Schwartz analytic square-torus (scattering off). Do not start T_{n+1} until T_n passes.

4 The adjoint / importance / contributon method

This is the conceptual heart of the magnet-shielding side and the part most worth internalizing.

4.1 Forward, adjoint, and reciprocity

Let L be the transport operator, q the source (neutrons born in the plasma), and ϕ the forward flux: $L\phi = q$. A *response* (e.g. fast-neutron flux or kerma in coil k) is

$$\mathcal{R}_k = \langle \Sigma_k, \phi \rangle = \int \Sigma_k(x) \phi(x) dx, \quad (7)$$

where Σ_k is the response function supported on coil k and $x = (\mathbf{r}, E, \hat{\Omega})$. Define the *adjoint* (importance) field by driving the transport *backward* from the detector:

$$L^\dagger \psi_k^\dagger = \Sigma_k. \quad (8)$$

Then reciprocity gives

$$\mathcal{R}_k = \langle \Sigma_k, \phi \rangle = \langle L^\dagger \psi_k^\dagger, \phi \rangle = \langle \psi_k^\dagger, L\phi \rangle = \langle \psi_k^\dagger, q \rangle = \int q(x) \psi_k^\dagger(x) dx. \quad (9)$$

What $\psi_k^\dagger$ means. $\psi_k^\dagger(x)$ is the *expected contribution to $\mathcal{R}_k$ of one particle introduced at phase-space point x* — the *importance* of x to coil k . Equation (9) says the coil load is the plasma source folded against importance. Solve the adjoint *once* (we use OpenMC’s random-ray solver, adjoint source = Σ_k) and you get importance *everywhere* on a mesh.

4.2 The contributon

Define the (angle-integrated, scalar) *contributon density*

$$C_k(x) = \phi(x) \psi_k^\dagger(x). \quad (10)$$

It is the density of the “flow” of particles that both *get born* (via ϕ tracing back to q) and *reach* coil k (via $\psi_k^\dagger$). Its streamlines trace the corridor from plasma to coil. Two projections of the *same* field answer two different questions.

4.3 Attribution vs. worth — the same field, different regions

Attribution. Restrict the reciprocity integrand $q \psi_k^\dagger$ to the *plasma*: it maps *where the neutrons responsible for coil k ’s load are born*. This is “source attribution.”

Worth. Add a thin absorber $\delta\Sigma$ at a *shield* location; first-order transport perturbation theory gives

$$\delta\mathcal{R}_k \approx - \int \delta\Sigma(x) \phi(x) \psi_k^\dagger(x) dx = - \int \delta\Sigma(x) C_k(x) dx. \quad (11)$$

So the *shielding worth* of adding material at x — the actionable design quantity $W_k(x) = -\partial\mathcal{R}_k/\partial m$ — is proportional to the contributon C_k *in the shield*.

One field, two regions.

$$\underbrace{q \psi_k^\dagger \text{ in the plasma}}_{\text{attribution: where born}} \quad \text{vs.} \quad \underbrace{\phi \psi_k^\dagger \text{ in the shield}}_{\text{worth: where blocking helps}}.$$

They differ because transport *carries* the contributon from plasma to shield along streaming paths — so the hottest birth region and the best shield patch need not line up. That gap is the whole experiment.

4.4 The three-way distinction (do not conflate)

Quantity	Formula / meaning	Purpose
Attribution	$C_k = q \psi_k^\dagger$ in plasma; where responsible neutrons are born	Diagnostic — explains the load
VR importance	$\psi_k^\dagger$ used to bias transport (CADIS/FW-CADIS weight windows)	Compute the deep response cheaply — <i>not</i> a design lever
Shielding worth	$W_k = -\partial \mathcal{R}_k / \partial m \propto \phi \psi_k^\dagger$ in shield	Actionable — what to build where

Table 2: Three uses of the adjoint that share $\psi_k^\dagger$ but answer different questions. Conflating them is the biggest reviewer risk; the caveat below matters.

Scalar-worth caveat. $\phi \psi^\dagger$ is the *leading-order scalar* worth. Exact anisotropic worth needs angular flux moments (an angular random-ray tally). So W is a heuristic we *validate against finite difference*, not a quantity we claim is exact. This is precisely why Tier 2 (rebuild geometry with real added shield, re-tally, difference) exists.

5 The stellarator neutronics computation stack

- **Configurations.** QH (Landreman–Paul quasi-helical, $n_{\text{fp}} = 4$, 20 coils) is the workhorse; QA (quasi-axisymmetric) as a cross-check. Plasma boundary from a VMEC `wout`.
- **Geometry.** **ParaStell** lofts an 8-layer radial build outward from the plasma and writes a watertight **DAGMC** `.h5m`. Layers (cm): first wall 3.2, Be multiplier 2, breeder 50, back wall 4, shield 40, gap 2, vacuum vessel 25, thermal shield 3. Coils are added as filament-swept solids.
- **Source.** **StellaratorSource** (§3) with a fluxmap field model supplies the DT birth sites and (optionally polarized) directions.
- **Response.** Per-coil fast-flux / kerma tallies (cells 11–30 for QH’s 20 coils).

5.1 The design variable: fixed-envelope breeder-for-shield trade

On a per-field-period 13×19 ($\phi \times \theta$) grid, a *delta* field re-partitions material *without moving the coils or growing the machine*:

$$t_{\text{shield}} = T_{\text{SH0}} + \delta, \quad t_{\text{breeder}} = T_{\text{BR0}} - \delta, \quad t_{\text{shield}} + t_{\text{breeder}} = \text{const (envelope conserved)}, \quad (12)$$

with a breeder floor ($t_{\text{breeder}} \geq 15$ cm) so tritium breeding is not destroyed. δ is stellarator-symmetric: defined per period, repeated n_{fp} times. Attribution and worth must therefore be measured in the *period frame* ($\phi \bmod 2\pi/n_{\text{fp}}$) to match the design knob.

Weight-window lesson (hard-won). FW-CADIS weight windows *over-split* on this geometry: 2M source particles spawned 6.8×10^7 secondary tracks and batch 1 never finished. The fix is brute-force *no*-WW sampling (the deep coil dose still resolves to 0.5–2.4% relative error at ~ 48 M histories). Matching settings across runs also makes them *correlated* (shared seed), which shrinks finite-difference variance for free.

6 Recent developments and results

6.1 Kill-shot: informed placement works, but the peak migrates

Using the coil-20 attribution map to place a nonuniform shield (fixed envelope) versus a uniform reference, at matched settings:

Metric	placed/uniform	reduction	note
Peak coil fast flux	0.240	4.2 $\times$	uniform peak = coil 20
Coil-20 (adjoint target)	0.215	4.7 $\times$	
Mean over coils	0.441	2.3 $\times$	

At fixed envelope the informed shield cuts peak coil flux to 24% of uniform. **But the peak migrated coil 20 $\rightarrow$ coil 25:** single-coil attribution over-protects its own target and leaves a different, unprotected coil as the new binding constraint. This is direct evidence that attribution is not, by itself, the right design lever — and it motivates both multi-coil importance and the patch-worth test.

6.2 Multi-coil importance (zero new transport)

All 20 per-coil adjoint maps already exist and share one mesh, so a multi-coil importance $\psi_{\text{multi}}^\dagger$ is pure post-processing: $\sum_k \psi_k^\dagger$ (total dose), $\max_k \psi_k^\dagger$ (envelope), or flux-weighted $\sum_k R_k \psi_k^\dagger$ (peak-driven). The flux-weighted map reveals the uniform hot region is the coil 17–20 *cluster* with coil 25 secondary — explaining the migration. This feeds the *existing* placement allocator unchanged; only the field it consumes changes.

6.3 Patch-worth: the decisive experiment

Partition the shield into ~ 16 (θ, ϕ) patches (period frame). For each, compute attribution $A = q\psi^\dagger$ (plasma) and scalar worth $W = \phi\psi^\dagger$ (shield band), and rank-correlate.

Tier 1 (done). A and W are *near-independent*, robustly across binnings:

grid	Spearman ρ	p	Chatterjee ξ
4×4	+0.26	0.33	0.15
5×5	+0.11	0.61	0.01
6×6	+0.13	0.46	0.004

Underpowered per binning (16–36 patches), but the *consistency* plus $\xi \approx 0$ points one way: **where neutrons are born $\neq$ where blocking helps.**

Tier 2 (in progress). Finite-difference validation: pick disagreement corners — e.g. patch (2, 1) has near-zero A but the *highest* W ; (2, 3) the highest A but modest W — rebuild each with +20 cm real shield, re-tally coil-20 dose, and difference against the (correlated) baseline to get $W_{\text{FD}} = -\Delta\mathcal{R}/\Delta m$. The verdict:

- $W_{\text{FD}} \sim W$ and $W_{\text{FD}} \not\sim A \Rightarrow$ *attribution is not the lever; worth is* (the novel result).
- $W_{\text{FD}} \sim W \sim A \Rightarrow$ attribution is a validated cheap proxy.
- $W_{\text{FD}} \not\sim W \Rightarrow$ scalar worth inadequate; need angular moments.

Engineering lesson from Tier 2. Concurrent doses must run in *separate working directories*: OpenMC writes a default `statepoint.N.h5`, so two jobs sharing a cwd clobber each other and return bit-identical (wrong) results. Isolate the cwd per task; key success on the output file, not the exit code (a benign teardown double-free can exit non-zero after a valid write).

7 Direction, scope, and novelty

7.1 Prior art (what is already taken)

- **Bae et al. (2025)** — polarized source in OpenMC. Takes the “SPF-source-in-OpenMC” flag; so Paper A’s novelty is the *coupling to magnet importance*, not the source.
- **Bogaarts & Warmer (2026)** — differentiable stellarator geometry + deterministic transport + FW-CADIS + nonuniform blanket optimization. Owns “nonuniform stellarator shield optimizer”; do *not* claim that phrase. But it never tests attribution $\neq$ worth and never targets magnets at fixed envelope.
- **ParaStell (Moreno 2024)** — manual radial-build sweep; names automated optimization as future work (the door we walk through).
- **CFS IAEA (2025)** — contribution/adjoint, but for a *diagnostic detector*, deterministic, tokamak. Not a magnet, not a stellarator.

7.2 The defensible claim

Compete on the physics, not the tool. The importance map everyone treats as a placement lever (attribution / VR importance) is *not* the shielding worth. We demonstrate the gap (Tier 1), validate the worth against finite difference (Tier 2), and use it to fix the peak-migration failure (multi-coil placement). Paper A adds the orthogonal SPF-directionality $\times$ magnet-importance coupling. None of the prior art performs this experiment.

7.3 Outlook

(1) Finish Tier 2 $\rightarrow$ the W_{FD} vs W vs A figure (the paper’s spine). (2) Feed the winning field (worth, or validated-attribution) to the existing allocator, now *multi-coil*, and show the migrated peak is fixed with a second-round reduction. (3) Cross with the SPF source for Paper A. Downstream, our objective-correction is exactly what a differentiable optimizer (à la Bogaarts & Warmer) should consume.

8 Symbol & concept cheat-sheet

a, b, c	Collision-mode fractions, $a + b + c = 1$; unpolarized = $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. $a=A$ (aligned $\perp$ lobe), b, c share the $\parallel$ lobe.
θ	Angle between neutron direction $\hat{\mathbf{u}}$ and local field $\hat{\mathbf{B}}$.
σ_0	$S = 3/2$ combined-spin cross section; $\sigma_{\text{tot}} = \sigma_0(a + \frac{2}{3}b + \frac{1}{3}c)$.
$w_{\perp}, w_{\parallel}$	Angular weights $\frac{3}{4}a$ and $\frac{2}{3}b + \frac{1}{3}c$; set the birth- <i>direction</i> pdf.
η	Total-rate factor $a + \frac{2}{3}b + \frac{1}{3}c$; scales source <i>strength</i> , separate from direction.
q	Forward source (neutron births in plasma). $L\phi = q$.
ϕ	Forward flux; $\mathcal{R}_k = \langle \Sigma_k, \phi \rangle$ is coil- k load.
$\psi_k^\dagger$	Adjoint/importance for coil k : $L^\dagger \psi_k^\dagger = \Sigma_k$; $\mathcal{R}_k = \langle q, \psi_k^\dagger \rangle$.
C_k	Contribution $\phi \psi_k^\dagger$: flow of particles born <i>and</i> reaching coil k .
Attribution A	$q \psi_k^\dagger$ in the <i>plasma</i> — where responsible neutrons are born.
Worth W	$-\partial \mathcal{R}_k / \partial m \propto \phi \psi_k^\dagger$ in the <i>shield</i> — where blocking helps.
W_{FD}	Finite-difference worth $-\Delta \mathcal{R}_k / \Delta m$ from a real rebuilt-and-retallied patch.
δ	Per-period shield-thickness increment traded against breeder at fixed envelope.
n_{fp}	Field periods (QH = 4); fold $\phi \bmod 2\pi/n_{\text{fp}}$ to match the design knob.

Physics per Schwartz (2025), Eqs. 1–2, re-derived here. Numbers from the QH corrected 8-layer build; Tier-2 finite difference in progress.